

Toan Vo

✉ toanvongoduc2004@gmail.com | [toannvo](#) | ☎ 608-381-8510 | [toanvond](#) | Portfolio: toanvond.github.io

PROFESSIONAL SUMMARY

Software Developer and Machine Learning enthusiast with 5 years of experience in software development and 4 years of hands-on experience in machine learning. Proficient in software engineering with strong expertise in Python, C/C++, and JavaScript. In-depth knowledge of modern software frameworks such as NodeJS and ReactJS. Experienced in developing machine learning models, with a focus on Deep Learning and Data Analysis. Skilled in popular ML frameworks such as scikit-learn, PyTorch, TensorFlow, and JAX, with a growing interest in foundational AI, ML interpretability and its applications in real-world problems.

EDUCATION

University of Maryland - College Park

M.S., Computer Science

College Park, MD

August 2026 – Present

- **Relevant courses:** Graduate NLP, Parallel Computing

University of Wisconsin - Madison

Madison, WI

B.S., Computer Science. Certificate: Statistics, Mathematics. GPA: 3.7/4

August 2022 – May 2026

- **Relevant courses:** Algorithms, Data Structures, Statistical Modeling, Object-Oriented Programming, Operating Systems, Database Systems, Probability Theory, Linear Algebra, Optimization, Deep Learning

EXPERIENCE

Software Engineer Intern

June 2026 – August 2026

A Story Technologies, Inc.

Remote (Vietnam)

- Developed core features of a cross-platform **React Native/Expo** app for AI-guided storytelling, piloted with **30+** users at retirement communities.
- Built a **Stripe** subscription billing system with hosted checkout, webhook-driven plan state, and self-serve cancel/resume, backed by **Supabase**.
- Engineered a multi-table **Postgres** schema with ownership-chained **Row-Level Security** and folder-scoped **Storage** bucket policies.
- Built a **Flask API** integrating **OpenAI API** and **Supabase** for AI chat/voice, with **JWT** auth (**JWKS**), usage logging, and signed-URL file storage.
- Engineered a **WebRTC** real-time voice AI pipeline with multimodal chat and token-level cost accounting.
- Architected stateless backend infrastructure on **AWS ECS/Fargate** with **ECR**-managed containers and centralized secrets management.

Software Engineer Intern

June 2025 – August 2025

Viettel Solutions

Ho Chi Minh City, Vietnam

- Designed and containerized AI model evaluation environments using **Docker**, standardizing workflows and ensuring reproducibility across platforms.
- Implemented local **LLM** inference pipelines via **Ollama**, enabling rapid model testing and benchmarking without dedicated GPU infrastructure.
- Optimized Viettel’s AI Tessel **computer vision** pipeline for multi-angle grocery shelf image analysis, increasing matching accuracy by 30% and reducing end-to-end processing time by 25%.

PROJECTS

MiniSpark | C

April 2025

- Built MiniSpark, a single-node implementation of Apache Spark in C, supporting RDD DAGs (Resilient Distributed Datasets) with transformations (map, filter, join, partitionBy) and actions (count, print).
- Implemented a thread pool scheduler with locks and condition variables to materialize RDD partitions in parallel, added metrics logging for profiling, and ensured safe memory management.

Workspace Manager | *MongoDB, Javascript*

November 2023

- Developed an application to streamline workspace setup and enhance workflow efficiency for multiple users.
- Implemented MongoDB-based backend to manage settings and preferences, reducing setup time by 50%.
- Incorporated secure authentication and encrypted accounts, supporting unlimited user customization.

SAT Score Analysis and Attendance Impact | R

October 2022 – December 2022

- Conducted statistical analysis to determine correlation between school attendance and SAT performance across New York high schools.
- Utilized R for data cleaning, processing, and hypothesis testing, identifying significant differences in reading and writing scores (mean difference: 6.54 points).

Highway Crossing 3D Game | OpenGL, C#

January 2022 – May 2022

- Designed and implemented a 3D highway crossing simulation with audio effects and textured 3D models for immersive gameplay.
- Optimized collision detection and OpenGL rendering pipeline, reducing latency by 20% and improving responsiveness.

TECHNICAL SKILLS

- **Programming & Data Analysis:** Python, Java, C/C++, SQL, JavaScript, TypeScript, Bash.
- **Tools & Platforms:** Git, GitHub, Supabase, Docker, AWS, Stripe, OpenAI API, REST APIs, EAS Build.
- **Libraries & Frameworks:** React Native, Expo, React Navigation, Flask.

HONORS AND AWARDS

- Finalist - **Madhacks Fall 2023** - Selected among top teams for innovative software solutions in competitive hackathon environment.
- Finalist - **MadData Fall 2024** - Recognized for excellence in data analysis and machine learning application.